

NAME: NEPTUN-CODE: TUTOR:

Probability Theory 1 Retaken 1st midterm, 22nd October 2024.

16:15–17:00

Working time: 45 minutes. Only simple scientific, non-programmable calculators are allowed.

Maximum score (with the bonus exercise): 24 points, but we consider 20 points already as 100%.

1. The Random Cola company introduces the following contest: In every cap of the bottles, they write one of the letters "W", "I" and "N" uniformly randomly. The letters in each cap are chosen independently. Whoever can form the word WIN (that is, collects three caps with all three letters) wins an excursion to Random City. I bought six bottles.
 - (a) What is the probability that I found at least one "W"? (3 points)
 - (b) What is the probability that I won exactly two bottles (That is, I found exactly 2 "W", 2 "I" and 2 "N")? (3 points)
 - (c) What is the probability that I did not win an excursion (That is, I did not collect all three letters)? (4 points)

Bonus: What is the expected number of collected different letters? (4 points)

2. We deal all cards from a well shuffled French deck (52 cards). Each symbol is associated to points as follows: Ace is 1 point, Jack is 11 points, Queen is 12 points and King is 13 points. The numbered cards worth as many points as their number. We pick two cards randomly.
 - (a) What is the probability that they worth the same amount of points? (2 points)
 - (b) What is the probability that the worth of the first card is strictly larger than the second card? (3 points)
 - (c) The sum of the values of the two cards is 6. What is the probability that the first card is an Ace? (5 points)